

Problem 1:

$$a) \frac{\partial J}{\partial \omega_1} = 6\omega_1 - 6 \quad \Rightarrow \quad \nabla J(\omega) = \begin{bmatrix} 6\omega_1 - 6 \\ 2\omega_2 - 4 \end{bmatrix}$$

$$\frac{\partial J}{\partial \omega_2} = 2\omega_2 - 4$$

$$b) \nabla J(\omega) = 0$$

$$6\omega_1 - 6 = 0 \rightarrow \omega_1^* = 1$$

$$2\omega_2 - 4 = 0 \rightarrow \omega_2^* = 2$$

$$\Rightarrow \omega^* = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$c) \text{ initial state: } \omega^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{initial gradient: } \nabla J(\omega^{(0)}) = \begin{bmatrix} -6 \\ -4 \end{bmatrix}$$

$$A) \alpha = 0.05$$

$$\omega_A^{(1)} = \omega^{(0)} - \alpha \nabla J(\omega^{(0)}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.05 \begin{bmatrix} -6 \\ -4 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.2 \end{bmatrix}$$

$$B) \omega_B^{(1)} = \omega^{(0)} - \alpha \nabla J(\omega^{(0)}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.12 \begin{bmatrix} -6 \\ -4 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix}$$

$$d) \|\omega_A^{(1)} - \omega^*\|_2 = \sqrt{(0.3-1)^2 + (0.2-2)^2} = 1.94$$

$$\|\omega_B^{(1)} - \omega^*\|_2 = \sqrt{(1.2-1)^2 + (0.8-2)^2} = 1.21$$

Candidate II converges faster towards the optimal point because of the larger learning rate. Although it slightly overshoots in the ω_1 dimension ($1.2 > 1$), the step size satisfies the quadratic stability condition ($|1 - \alpha \lambda_i| < 1$). Therefore, the oscillation is damped, and it rapidly approaches the minimum without diverging.

Problem 2 :

a)

Hypothesis: $\hat{y} = Xw + b$

MSE Cost Function: $J(w, b) = \frac{1}{2m} (Xw + b - y)^T (Xw + b - y)$

Gradient Update Rules: $w^{(new)} = w^{(old)} - \alpha \left[\frac{1}{m} X^T (\hat{y} - y) \right]$

$b^{(new)} = b^{(old)} - \alpha \left[\frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) \right]$

b) $w^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $b^{(0)} = 0$, $\alpha = 0.1$, $m = 3$

Epoch 1: $\hat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \hat{y} - y = \begin{bmatrix} -2 \\ -5 \\ -3 \end{bmatrix}$ $\left. \begin{array}{l} \nabla_w J = \begin{bmatrix} -4 \\ -2.66 \end{bmatrix}, \nabla_b J = -3.33 \end{array} \right\} \rightarrow \begin{array}{l} w^{(1)} = \begin{bmatrix} 0.4 \\ 0.266 \end{bmatrix} \\ b^{(1)} = 0.33 \end{array}$

Epoch 2: $\hat{y} = \begin{bmatrix} 0.7 \\ 1.4 \\ 0.6 \end{bmatrix} \rightarrow \hat{y} - y = \begin{bmatrix} -1.266 \\ -3.6 \\ -2.4 \end{bmatrix}$ $\left. \begin{array}{l} \nabla_w J = \begin{bmatrix} -2.82 \\ -2 \end{bmatrix}, \nabla_b J = -2.42 \end{array} \right\} \rightarrow \begin{array}{l} w^{(2)} = \begin{bmatrix} -0.1682 \\ 0.466 \end{bmatrix} \\ b^{(2)} = 0.5756 \end{array}$

c)

initial: $J_0 = \frac{1}{6} (4 + 25 + 9) = 6.33$

After Epoch 2: $\hat{y} = \begin{bmatrix} 1.2578 \\ 2.4067 \\ 1.0423 \end{bmatrix}$

$J_2 = \frac{1}{6} [(0.7422)^2 + (2.5933)^2 + (1.9577)^2] = 1.8515$

The cost decreases, so Gradient Descent successfully reduces the loss.